DAY 22 | PRE-READ

Giving AI the Right Information

A light read before Day 22. About 15 minutes, no code required.

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Why read this?

Yesterday you learned to write good instructions for AI. But even a perfect instruction fails if the AI does not have the facts it needs. Tomorrow is about choosing what information to give the model, and building systems that fetch the right information automatically.

1. The open-book exam

In a closed-book exam you rely on memory, and you might guess. In an open-book exam you look things up, and your answers are more accurate. An AI model on its own is in a closed-book exam: it only knows what it learned during training, which does not include your company's documents. **RAG** (Retrieval-Augmented Generation) turns it into an open-book exam: the system finds the relevant pages and hands them to the model with the question.

2. A desk with limited space

The model can only look at a limited amount of text at once, like a desk that holds only so many papers. **Context engineering** is deciding which papers go on the desk: the instructions, the question, the most relevant documents, and a short summary of the conversation so far. Too little and it lacks facts; too much and the important bits get lost.

> **Key idea:** the quality of an AI answer depends as much on the information you give it as on the model itself.

3. Cutting documents into useful pieces

You cannot hand the model a 300-page manual every time. Instead, documents are split into smaller passages called **chunks**, and only the most relevant chunks are used. Cutting well matters: a chunk that says "it expires on 31 March" is useless if the word "leave" ended up in a different chunk.

4. Two pipelines

Pipeline	When	Steps
Preparation (ingestion)	Ahead of time	Load documents, clean, cut into chunks, store
Answering (query)	Every question	Find relevant chunks, hand them to the model, answer with sources

5. Think about these before class

- Which documents at your workplace would be most valuable to make searchable by AI?
- If you had to cut your company's leave policy into short pieces, where would you cut?
- Why would citing the source of an answer increase trust?

Senpai (先輩): a senior colleague who guides you